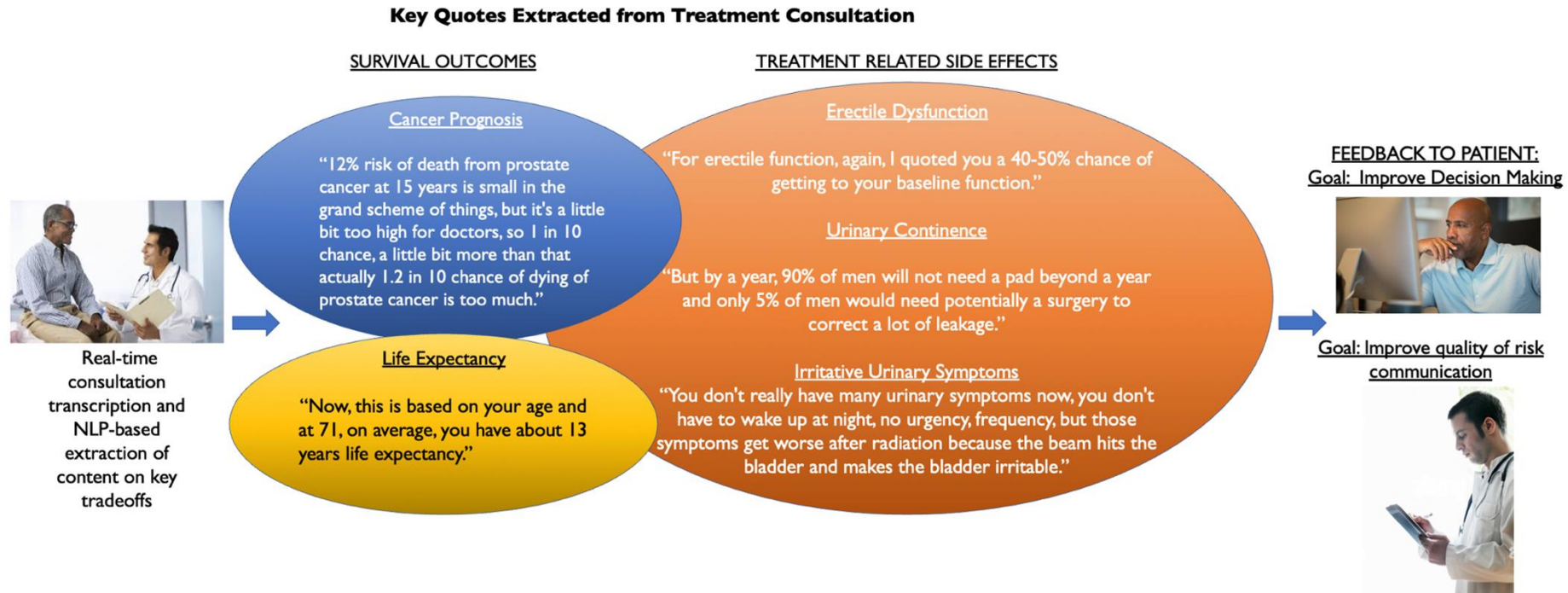


Patient and Physician Reports: Data Visualization Plan

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Cluster Randomized Trial of NLP+AI-Based Feedback from Prostate Cancer Consultations



Cluster randomized at level of physician to:

- NLP+AI summaries of key topics emailed after the visit
- Standard of care (AUA-published literature on risk communication (doctors) and decision making (pts))

Can real-time NLP+AI-based feedback on what was said in prostate cancer consultations improve quality of physician communication and patient decision making?

Patient Reports

Introduction

One of the challenges in navigating a prostate cancer diagnosis is making a choice on treatment. There are numerous treatment options—each with unique risks and benefits—which makes deciding between them difficult. An important goal of the treatment consultation for physicians is to clearly communicate the risks and benefits of treatment to help you make an informed treatment decision. However, since these consultations cover a lot of information and can last a long time, it is tough to remember the key facts when considering treatment.

To help address this problem, we created a program that captures key information from the treatment consultation to help you remember this information when considering your treatment options. The attached report identifies the key content from what your doctor discussed during your consultation about 5 important topics: cancer prognosis, life expectancy, and major treatment-related side effects (erectile dysfunction, urinary incontinence, and lower urinary tract symptoms). The report identifies the top sentences from your doctor about these topics. At the end of each section is also an artificial intelligence-generated summary of what was discussed on each topic. After reviewing this material, we will ask you to fill out a questionnaire to see how this information helps or does not help you in making your treatment decision. Later, we will conduct an interview to determine how this system can be improved to meet your needs.

Life Expectancy

- like i said, you've got 40 years ahead of you.
- so that's a good thing for a patient who has a lot of years ahead of them.
- but for a person like you who is young and has, you know, you know, you've got 40 years ahead of you-
- but for you, having many years ahead of you, you have an intermediate-risk tumor that's kind of on the high end of the intermediate-risk scale.
- but personally, i think, you know, you're a young man, you've got a ton of years ahead of you, surgery gives you good local control, and it gives you the options for salvage therapy if you need it in the future.
- so you're about 20 years ahead of the game.
- but we want to plan for those 40 years.
- so we've got to plan for the long-term for you.
- and the likelihood it would threaten your life over that 40-year period, i think we're getting up towards 40%.

Summary

- Based on your situation, you have a tumor that is considered intermediate-risk, which is on the higher end of that scale. However, considering your young age and the fact that you have many years ahead of you, surgery would provide you with good local control and the option for additional treatment if needed in the future. Although the likelihood of it potentially threatening your life over the next 40 years is around 40%, we need to plan for the long-term and ensure that we take all necessary measures to address this.

Patient Report

- Introductory information explaining what results are (ie slide 4)
- For each of 5 domains (cancer prognosis, life expectancy, erectile dysfunction, urinary incontinence, and irritative lower urinary tract symptoms):
 - Top 5 sentences extracted by NLP
 - AI summary (will send scripts and prompts for this)

Physician Reports

Physician Report

- One page summarizing results in grid format (ie slide 9)
- For each of 5 domains (cancer prognosis, life expectancy, erectile dysfunction, urinary incontinence, and irritative lower urinary tract symptoms) a separate “page” showing a display like slide 10.
 - However, the group felt like the x and y axes in the figure should be switched, making each level of quality vertical, with highest level on top
 - Suggestions for improvement (specific to each domain) are on slides 11-16
 - Only display suggestions for improvement that are higher than the scored sentence (ie if score was 3, only show suggestions to achieve score of 4 and 5)
 - Also would want the capability for doctors to provide a new sentence that would then be graded again by our AI scoring – could be verbal or type written

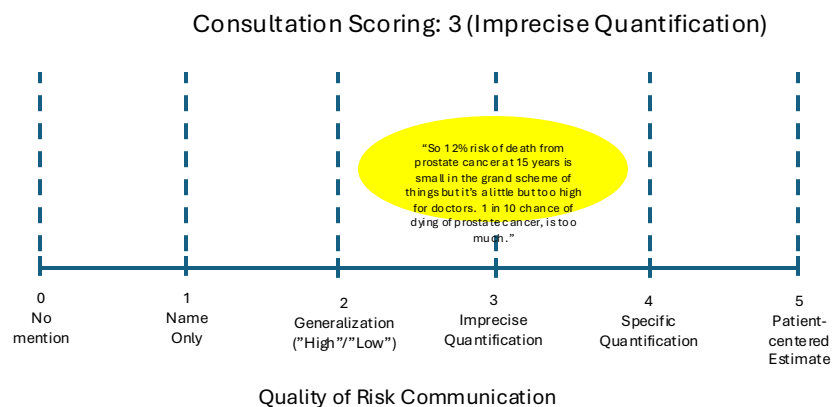
Grid summary

Topic	Your Score	Representative Sentence	Suggested Rephrasing
Cancer Prognosis		Lorem ipsum dolor sit amet, consectetur flibbertonium elit, sed do wibber magna aliqua.	None suggested
Life Expectancy		Lorem ipsum dolor sit amet, consectetur flibbertonium elit, sed do wibber magna aliqua.	<u>Suggestion:</u> Add info on X, Y, Z Lorem ipsum dolor sit amet, consectetur flibbertonium elit, sed do wibber magna aliqua.
Urinary Incontinence		Lorem ipsum dolor sit amet, consectetur flibbertonium elit, sed do wibber magna.	<u>Suggestion:</u> Add info on X, Y, Z Lorem ipsum dolor sit amet, consectetur flibbertonium elit, sed do wibber magna aliqua.
Erectile Dysfunction		Lorem ipsum dolor sit amet, consectetur flibbertonium elit, sed do wibber magna.	<u>Suggestion:</u> Add info on X, Y, Z Lorem ipsum dolor sit amet, consectetur flibbertonium elit, sed do wibber magna.

Score should be color coded and numeric 1 red 2 pink 3 yellow 4 ?light green 5 green;
symbols unnecessary – should be colors and numbers

This should also have an additional column for suggestions for improvement in
between the third and fourth column

Cancer Prognosis



Suggestions for Improvement

- Provide cancer prognosis with and without treatment (to achieve score of 4)
- Provide cancer prognosis at life expectancy (to achieve score of 5)

AI Re-write

“So 12% of risk of death from prostate cancer at your life expectancy of 17 years is a little too high. Treatment will reduce that risk to 6%.”

Full Context

“I hope you're not going to die of your cancer, I want to prevent that. So of course everyone who gets diagnosed with cancer wants to know what is the risk that the cancer will harm them. We know this data from studies looking at risk of cancer death in men who chose not to treat their cancers. In your case, that risk turns out to be 12% at 15 years. **So 12% risk of death from prostate cancer at 15 years is small in the grand scheme of things, but it's a little bit too high for doctors, so 1 in 10 chance, a little bit more than that actually 1.2 in 10 chance of dying of prostate cancer is too much.** So how do we get that number lower? We would treat with surgery or radiation. But would it be unreasonable to just watch this? Not totally. It's still on the table that you could potentially just watch this, but it's a value judgment, and we could match kind of what you think is important with your goals. But for the majority of these unfavorable risks, intermediate risk prostate cancer that I do recommend treatment, just to be on the safe side.”

Suggestions for Improvement

Suggestions for Improvement: Cancer Prognosis

- Discuss potential for risk of cancer death, metastasis, or progression (to achieve score of 1)
- Provide a generalization of magnitude of risk (“high”/”low”) (to achieve a score of 2)
- Provide a quantified estimate of risk (to achieve a score of 3)
- Provide a quantified estimate of risk both with treatment and without treatment at an arbitrary timepoint (to achieve score of 4)
- Provide a quantified estimates of risk both with and without treatment at the patient’s life expectancy (to achieve score of 5)

Suggestions for Improvement: Life Expectancy

- Discuss the concept of competing risks of mortality (to achieve score of 1)
- Provide a generalization of duration of life expectancy (i.e., “long”/”short”) (to achieve a score of 2)
- Provide a rough quantified estimate of life expectancy (e.g., “about 15-20 years”) (to achieve a score of 3)
- Provide a probability of living to an arbitrary timepoint (to achieve score of 4)
- Provide a specific number of years and mention calculation based on the patient’s age and health status (to achieve score of 5)

Suggestions for Improvement: Erectile Dysfunction

- Discuss the potential risk of erectile dysfunction (to achieve score of 1)
- Provide a generalization of risk (i.e., “high”/”low”) (to achieve a score of 2)
- Provide an average probability of ED without a time horizon (e.g., “45% risk of erectile dysfunction”) (to achieve a score of 3)
- Provide an average probability of ED with a time horizon (e.g., “45% risk of erectile dysfunction at 1 year postop”) (to achieve a score of 4)
- Provide a patient-specific probability of ED with a time horizon, mentioning patient-specific factors such as age and baseline function that inform your estimate (to achieve score of 5)

Suggestions for Improvement: Urinary Incontinence

- Discuss the potential risk of urinary incontinence (to achieve score of 1)
- Provide a generalization of risk (i.e., “high”/”low”) (to achieve a score of 2)
- Provide an average probability of UI without a time horizon (e.g., “10% risk of needing pads”) (to achieve a score of 3)
- Provide an average probability of UI with a time horizon (e.g., “10% risk of needing pads beyond 1 year postop”) (to achieve a score of 4)
- Provide a patient-specific probability of UI with a time horizon, mentioning patient-specific factors such as age and baseline function that inform your estimate (to achieve score of 5)

Suggestions for Improvement: Irritative Urinary Symptoms

- Discuss the potential risk of irritative urinary symptoms (to achieve score of 1)
- Provide a generalization of risk (i.e., “high”/”low”) (to achieve a score of 2)
- Provide an average probability of LUTS without a time horizon (e.g., “30% risk of developing irritative urinary symptoms”) (to achieve a score of 3)
- Provide an average probability of LUTS with a time horizon (e.g., “30% risk of developing irritative urinary symptoms that may or may not resolve over the following year”) (to achieve a score of 4)
- Provide a patient-specific probability of LUTS with a time horizon, mentioning patient-specific factors such as age and baseline function that inform your estimate (to achieve score of 5)